

A Feel for Dividing by $|\nabla\rho|$

This note carries no proofs. It is only trying to make one operation — dividing the force by $|\nabla\rho|$ — feel obvious. The proofs are in the companion notes.

1 The one-sentence version

ρ is a risk coordinate, not a length. Measuring “risk change per metre” quietly mixes in how many metres a unit of risk happens to cost in that direction — and that exchange rate varies by a factor of q around the ellipse. Dividing by $|\nabla\rho|$ pays in a single currency.

Everything below is an attempt to make that sentence land.

2 The demonstration

Stand at a point 0.5 m to the side of the obstacle and 9 m ahead of it, with $t_x = 1$, $t_y = 15$. Walk one metre in each of eight directions. Ask two questions each time: *how much did U change per metre?* and *how much did U change per unit of ρ ?*

heading	dU/ds (per metre)	$d\rho/ds$ (per metre)	$dU/d\rho$
0°	−10.49483	0.640184	−16.393443
45°	−8.01464	0.488893	−16.393443
90°	−0.83959	0.051215	−16.393443
135°	6.82729	−0.416464	−16.393443
180°	10.49483	−0.640184	−16.393443
225°	8.01464	−0.488893	−16.393443
270°	0.83959	−0.051215	−16.393443
315°	−6.82729	0.416464	−16.393443

Look at the middle column. Measured *per metre*, the field looks 12.5 times steeper if you walk sideways (0°) than if you walk straight ahead (90°). That asymmetry is the dip, in miniature.

Now look at the right column. Measured *per unit of ρ* , every single heading returns the same number, to six decimal places — and that number is exactly $U'(\rho) = -\lambda/\rho^2 = -16.393443$.

So the anisotropy was never in U . U has no opinion about direction at all; it only ever knew about ρ . The entire directional structure of $|\nabla U|$ was contributed by the conversion from ρ to metres. Divide that out and it vanishes, because there was nothing underneath it.

3 Why: the exchange rate

Think of ρ as a currency and metres as another. The field’s rule is simple and direction-blind: *risk depends on how many ρ you are from the obstacle*. But we do not read the field in ρ ; we read it in metres, and the exchange rate is terrible in one direction and fine in the other:

walking. . .	metres per unit of ρ	$ \nabla\rho $
sideways (minor axis)	$t_x = 1$	1
straight ahead (major axis)	$t_y = 15$	$1/15 = 0.0667$

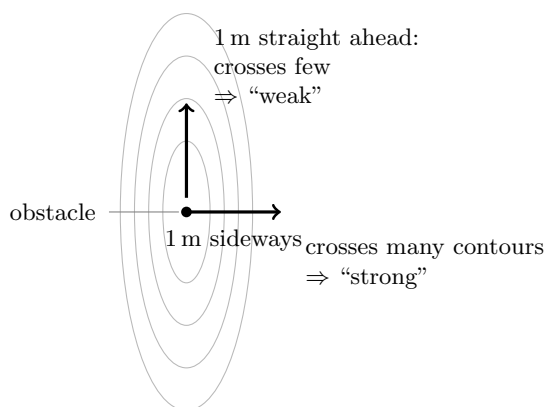
Walking straight ahead, one metre buys you only $1/15$ of a unit of ρ , so of course U barely moves — you have hardly gone anywhere *in the units the field actually cares about*. Reporting that as “weak force” is like reporting that a currency is weak without saying which one you are counting in.

$|\nabla\rho|$ is the exchange rate. Dividing by it converts the answer back into the currency the field was written in.

4 Why: the stretched map

A second way in, for those who prefer pictures. On a Mercator map, Greenland is drawn enormous. If you measured “temperature change per centimetre of paper” near the pole, you would conclude the temperature gradient there is tiny — not because the Arctic is thermally bland, but because the map stretches that region across far more paper than it deserves. To recover the real gradient you divide by the local map scale factor.

Our field is a map stretched $15:1$ in the longitudinal direction, and $|\nabla\rho|$ is precisely its local scale factor. The dip ahead of the obstacle is our Greenland.



Same one metre, same field, wildly different contour counts — entirely because of how the contours were drawn.

5 Where the dip actually comes from

The factorisation $|\nabla U| = |U'(\rho)| \cdot |\nabla\rho|$ says the force is a product of two things. Here they are, scanning sideways from the centreline at 9m ahead:

x	ρ	$ \nabla\rho $	$ U'(\rho) $	$ \nabla U $ (product)
0.00	0.6000	0.06667	27.7778	1.8519
0.10	0.6083	0.17706	27.0270	4.7855
0.20	0.6325	0.32249	25.0000	8.0623
0.30	0.6708	0.45117	22.2222	10.0260
0.44	0.7440	0.59380	18.0636	10.7262
0.60	0.8485	0.70868	13.8889	9.8427
1.00	1.1662	0.85818	7.3529	6.3101
2.00	2.0881	0.95802	2.2936	2.1973

Read the columns separately and the mystery dissolves:

- $|U'(\rho)|$ — the honest quantity — falls monotonically, $27.8 \rightarrow 2.3$. It never rises. It has no dip and no ridge.
- $|\nabla\rho|$ — the exchange rate — climbs monotonically, $0.067 \rightarrow 0.958$. This is the ruler un-stretching as you leave the stretched direction.
- Their product is **not** monotone. It rises to 10.73 at $x \approx 0.44$ and falls after. That peak is the ridge, and the low value at $x = 0$ is the dip.

The dip is not a property of the risk. It is what you get when a monotone decay is multiplied by a monotone climb and the climb wins first.

6 Why the climb wins first

One more layer, and this is the part worth internalising. Compare the first two rows. Moving just 0.10 m sideways:

	at $x = 0$	at $x = 0.10$	
ρ	0.6000	0.6083	(+1.4%)
$ \nabla\rho $	0.06667	0.17706	(+166%)

You have barely moved in risk terms — ρ changed by one percent — but the exchange rate has nearly tripled. That mismatch is the engine of the dip, and it has a clean cause: **the two quantities vary on different lateral scales**. Writing $a = |d_y|/t_y$ for the on-axis value of ρ ,

$$\rho \text{ changes over a lateral scale } \sim a, \quad |\nabla\rho| \text{ changes over a lateral scale } \sim a/t_y.$$

The exchange rate moves t_y times faster than the distance does. With $t_y = 15$ that is a fifteen-fold head start, so near the axis the climb always outruns the decay.

This also explains why the dip needs a *threshold* on q rather than appearing always. The climb has a hard ceiling: $|\nabla\rho|$ can rise from $1/q$ to 1 and no further — a total factor of q , and nothing more. Once it saturates near 1, the monotone decay of $|U'|$ takes over unopposed. If q is small there is simply not enough room in the climb to overturn the decay at all, and no dip ever forms. That is the origin of the $q > \sqrt{3}$ condition: it is the point at which the ceiling is high enough to matter.

7 What dividing does, then

Nothing clever. It deletes the second column.

$$\underbrace{|\nabla U|}_{\text{non-monotone}} = \underbrace{|U'(\rho)|}_{\text{monotone}} \times \underbrace{|\nabla \rho|}_{\text{the artefact}} \implies \frac{|\nabla U|}{|\nabla \rho|} = |U'(\rho)|.$$

What remains is the column that was always well behaved. Since $|U'(\rho)|$ depends on nothing but ρ , and ρ grows as you move away from the centreline in any direction, the result must decay outward. It cannot dip, at any elongation, because there is no longer a second factor available to push it back up.

8 And the direction is untouched

Worth saying plainly, because it is what makes this usable. $|\nabla \rho|$ is a *positive number* at every point. Dividing a vector by a positive number makes it shorter or longer; it cannot turn it. Every arrow in the quiver plot points exactly where it pointed before — only the colours change.

This is the whole difference between this fix and the metric-weighted alternative $-G^{-1}\nabla U$, which divides the two components by *different* numbers and therefore does swing the arrows round, flattening them onto the radial direction and throwing away the sideways push that makes the vehicle go around the obstacle rather than merely brake behind it.

9 The one thing you give up

An honest closing note. Because $|\nabla \rho|$ is not itself a function of ρ — it varies from $1/q$ to 1 around a single ellipse — the rescaled field is no longer the gradient of anything. There is no potential whose downhill direction it is. For a reactive controller this is usually fine, but it does mean U can no longer serve as a Lyapunov function, and any stability argument resting on that has to be rebuilt. The companion note quantifies the size of this effect.